

Jellyfish: Instruction-Tuning Local Large Language Models for Data Preprocessing

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Motivation & Problem

- Data preprocessing (DP) crucial but fragmented across tasks
 - transforming raw data into a clean, manageable, and processable format
 - DP involves disparate tasks
- LLM-based DP
 - internal knowledge, reasoning ability, and generalizability
 - often relies on GPT APIs: security and customization issues
 - Customization difficulty
- Goal: Local, instruction-tuned universal DP solver

Key Idea

- Instruction-tune open LLMs (7B–13B) for DP tasks
- Inject domain knowledge and reasoning data
- Models: Jellyfish-7B, 8B, 13B

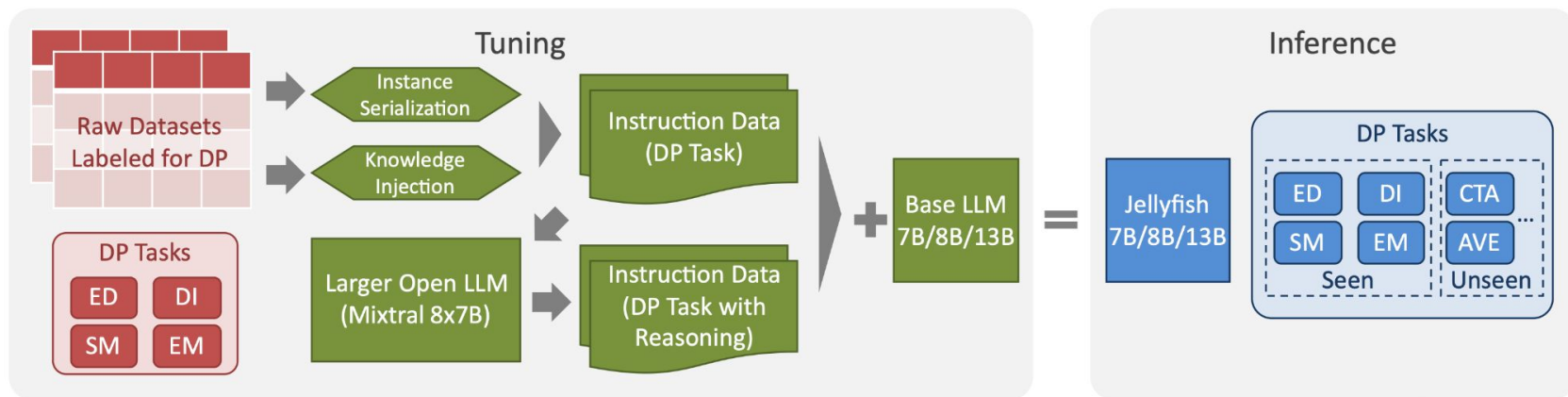


Figure 1: Overview of instruction tuning for data preprocessing.

Methodology

- Jellyfish tunes open LLMs
 - Mistral-7B (Jellyfish-7B), Llama 3-8B (Jellyfish-8B), and OpenOrca-Platypus2-13B (Jellyfish-13B)
- 4 seen tasks
 - Error Detection (ED)
 - Data Imputation (DI)
 - Schema Matching (SM)
 - Entity Matching (EM)
- 2 unseen tasks
 - Column Type Annotation (CTA)
 - Attribute Value Extraction (AVE)

Methodology

- The approach handles one record or a pair at a time (**instance-based processing**), making the prompt length manageable within LLM token limitations

Task	Dataset	#Instances	#Positives
ED	Adult	550×2	35×2
	Hospital	1710×2	44×2
DI	Buy	586	N/A
	Restaurant	778	N/A
SM	MIMIC-III	7000	11
	Synthea	5000	18
EM	Amazon-Google	6874	699
	Beer	359	54
	DBLP-ACM	5000	885
	DBLP-GoogleScholar	5000	924
	Fodors-Zagats	757	88
	iTunes-Amazon	430	105

#Positives denotes the number of instances having an error (for ED) or matching objects (for SM and EM)

×2 denotes duplication of instances for treating missing values as errors or not.

Instruction Data Construction Techniques

- Instance Serialization: Raw data is converted into **instruction tuning prompts** (e.g., system message, task description, instance content, question, and output format)
- Knowledge Injection: Infusing essential domain knowledge to guide the model, becomes built-in knowledge
 - General Knowledge (e.g. handling missing values)
 - Specific Knowledge (e.g. directing the model to prioritize certain attributes)

Instruction Data Construction Techniques

- Reasoning Data Distillation
 - Reasoning data is generated using a larger open LLM, Mixtral-8x7B-Instruct-v0.1
 - Distilling Mixtral's knowledge into Jellyfish
 - Ground truth
 - Answer hint is added to the prompt
- transform raw data to (1) DP task data, for DP task-solving ability, and (2) DP task with reasoning data, for interpretation ability.

Example Prompt

	DP Task Data	DP Task with Reasoning Data
system message	You are an AI assistant that follows instruction extremely well. User will give you a question. Your task is to answer as faithfully as you can.	You are an AI assistant that follows instruction extremely well. User will give you a question. Your task is to answer as faithfully as you can. While answering, provide detailed explanation and justify your answer.
task description	You are tasked with determining whether two Products listed below are the same based on the information provided. Carefully compare all the attributes before making your decision.	
injected knowledge	Note that missing values (N/A or "nan") should not be used as a basis for your decision.	
instance content	Product A: [name: "Sequoia American Amber Ale", factory: "Wig And Pen"] Product B: [name: "Aarhus Cains Triple A American Amber Ale", factory: "Aarhus Bryghus"]	
question	Are Product A and Product B the same Product?	
output format	Choose your answer from: [Yes, No]	After your reasoning, finish your response in a separate line with and ONLY with your final answer. Choose your final answer from [Yes, No].
answer	No	... the names are not identical, with Product B having ... These are different factories ... they are not the same product. Final answer: No

The reasoning data sets varied in size (8k, 11k, 14k, 20k instances)

Dataset	Task				
	ED	DI	SM	EM	Total
reasoning-8k	3056	1364	2000	2000	8420
reasoning-11k	3056	1364	3500	3500	11420
reasoning-14k	3056	1364	5000	5000	14420
reasoning-20k	3056	1364	8600	7000	20020

Experiments

- The 7B and 8B models are tuned both on DP task and reasoning data
 - 15k reasoning instances for 7B
 - 8k for the 8B
 - So suitable for unseen tasks
- 13B model is tuned with only DP task data, no reasoning
 - So task solver dedicated to the tuned tasks.

Test Data

Task	Type	Dataset	#Instances
ED	Seen	Adult	9900
		Hospital	17101
	Unseen	Flights Rayyan	12832 8997
DI	Seen	Buy Restaurant	65 86
		Flipkart Phone	2675 1194
	Unseen		
SM	Seen	MIMIC-III Synthea	6408 2964
	Unseen	CMS	2564
EM	Seen	Amazon-Google	2293
		Beer	91
		DBLP-ACM	2473
		DBLP-GoogleScholar	5742
		Fodors-Zagats	189
		iTunes-Amazon	109
	Unseen	Abt-Buy Walmart-Amazon	1946 2049
CTA	Unseen	SOTAB	250
AVE	Unseen	AE-110K	1482
		OA-Mine	2451

Results on Seen Tasks

- Jellyfish-13B the best average score (86.02) compared to GPTs
- Outperformed non-LLM methods on all seen datasets

Task	Type	Dataset	Model								
			Best of non-LLM	GPT-3	GPT-3.5	GPT-4	GPT-4o	Table-GPT	Jellyfish-7B	Jellyfish-8B	Jellyfish-13B
ED	Seen	Adult Hospital	<u>99.10</u> 94.40	<u>99.10</u> 97.80	92.01 90.74	92.01 90.74	83.58 44.76	– –	77.40 94.51	73.74 93.40	99.33 95.59
	Unseen	Flights Rayyan	81.00 79.00	– –	– –	83.48 <u>81.95</u>	66.01 68.53	– –	69.15 75.07	66.21 81.06	<u>82.52</u> 90.65
DI	Seen	Buy Restaurant	96.50 77.20	98.50 88.40	98.46 <u>94.19</u>	100 97.67	100 90.70	– –	98.46 89.53	98.46 87.21	100 89.53
	Unseen	Flipkart Phone	68.00 86.70	– –	– –	89.94 90.79	83.20 86.78	– –	87.14 86.52	<u>87.48</u> 85.68	81.68 <u>87.21</u>
SM	Seen	MIMIC-III Synthea	20.00 38.50	– 45.20	– <u>57.14</u>	40.00 66.67	29.41 6.56	– –	53.33 55.56	<u>45.45</u> 47.06	40.00 56.00
	Unseen	CMS	<u>50.00</u>	–	–	19.35	22.22	–	42.86	38.10	59.29
EM	Seen	Amazon-Google Beer	75.58 94.37	63.50 100	66.50 96.30	74.21 100	70.91 90.32	70.10 96.30	81.69 100.00	<u>81.42</u> 100.00	81.34 96.77
		DBLP-ACM	98.99	96.60	96.99	97.44	95.87	93.80	98.65	98.77	<u>98.98</u>
		DBLP-GoogleScholar	<u>95.70</u>	83.80	76.12	91.87	90.45	92.40	94.88	95.03	98.51
		Fodors-Zagats	100	100	100	100	93.62	100	100	100	100
		iTunes-Amazon	97.06	<u>98.20</u>	96.40	100	98.18	94.30	96.30	96.30	98.11
	Unseen	Abt-Buy	89.33	–	–	92.77	78.73	–	86.06	88.84	<u>89.58</u>
		Walmart-Amazon	86.89	87.00	86.17	90.27	79.19	82.40	84.91	85.24	<u>89.42</u>
Average			80.44	-	-	<u>84.17</u>	72.58	-	82.74	81.55	86.02

Results on Unseen Tasks

- Jellyfish models shows competitiveness
 - 7B and 13B
- For AVE, all the Jellyfish models showcase strong generalizability

Task	Dataset	Model									
		RoBERTa (159 shots)	RoBERTa (356 shots)	Stable Beluga 2 70B	SOLAR 70B	GPT-3.5	GPT-4	GPT-4o	Jellyfish- 7B	Jellyfish- 8B	Jellyfish- 13B
CTA	SOTAB	79.20	89.73	–	–	<u>89.47</u>	91.55	65.06	83.00	76.33	82.00
AVE	AE-110k	–	–	52.10	49.20	61.30	55.50	55.77	56.09	<u>59.55</u>	58.12
	OA-Mine	–	–	50.80	55.20	<u>62.70</u>	68.90	60.20	51.98	59.22	55.96

Comparison with Base Models

Task	Type	Dataset	Model					
			Mistral-7B	Jellyfish-7B	Llama 3-8B	Jellyfish-8B	OOP2-13B	Jellyfish-13B
ED	Seen	Adult	20.66	77.40 (+56.74)	47.42	73.74 (+26.32)	61.53	99.33 (+37.80)
		Hospital	37.09	94.51 (+57.42)	52.51	93.40 (+40.89)	63.24	95.59 (+32.35)
	Unseen	Flights	28.07	69.15 (+41.08)	67.71	66.21 (-1.50)	73.01	82.52 (+9.51)
		Rayyan	22.86	75.07 (+52.21)	62.46	81.06 (+18.64)	89.37	90.65 (+1.28)
DI	Seen	Buy	76.92	98.46 (+21.54)	86.15	98.46 (+12.31)	89.23	100 (+10.77)
		Restaurant	18.75	89.53 (+70.78)	43.02	87.21 (+44.19)	81.40	89.53 (+8.13)
	Unseen	Flipkart	79.52	87.14 (+7.62)	66.50	87.48 (+20.98)	78.49	81.68 (+3.19)
		Phone	76.72	86.52 (+9.80)	82.16	85.68 (+3.52)	84.33	87.21 (+2.88)
SM	Seen	MIMIC-III	6.90	53.33 (+46.43)	14.81	45.45 (+30.64)	36.36	40 (+3.64)
		Synthea	26.67	55.56 (+28.89)	23.52	47.06 (+23.54)	22.22	56 (+33.78)
	Unseen	CMS	0	42.86 (+42.86)	27.02	38.10 (+11.08)	13.33	59.29 (+45.96)
EM	Seen	Amazon-Google	36.51	81.69 (+45.15)	60.67	81.42 (+20.75)	36.70	81.34 (+44.64)
		Beer	69.57	100 (+30.43)	88	100 (+12)	85.71	96.77 (+11.06)
		DBLP-ACM	85.30	98.65 (+13.35)	82.14	98.77 (+16.63)	78.86	98.98 (+20.12)
		DBLP-GoogleScholar	59.54	94.88 (+35.34)	76.15	95.03 (+18.88)	59.48	98.51 (+39.03)
		Fodors-Zagats	66.67	100 (+33.33)	95.23	100 (+4.77)	92.68	100 (+7.32)
		iTunes-Amazon	70.97	96.30 (+25.33)	79.36	96.30 (+16.94)	57.45	98.11 (+40.66)
	Unseen	Abt-Buy	36.99	86.06 (+49.07)	44.60	88.84 (+44.24)	31.51	89.58 (+58.07)
		Walmart-Amazon	63.14	84.91 (+21.77)	59.69	85.24 (+25.55)	65.21	89.42 (+24.21)
		CTA	Unseen	SOTAB	23.49	83.00 (+59.1)	64.25	76.33 (+12.08)
AVE	Unseen	AE-110k	32.92	56.09 (+23.17)	56.33	59.55 (+3.22)	43.87	58.12 (+14.25)
		OA-Mine	32.44	51.98 (+19.54)	55.57	59.22 (+3.65)	54.81	55.96 (+1.15)
Average			44.17	80.14 (+35.97)	60.69	79.30 (+18.60)	61.60	83.21 (+21.61)

Impact of Instruction Data Configuration

- Tune the 13B model with data for a single DP task
- Randomly sample data from the datasets

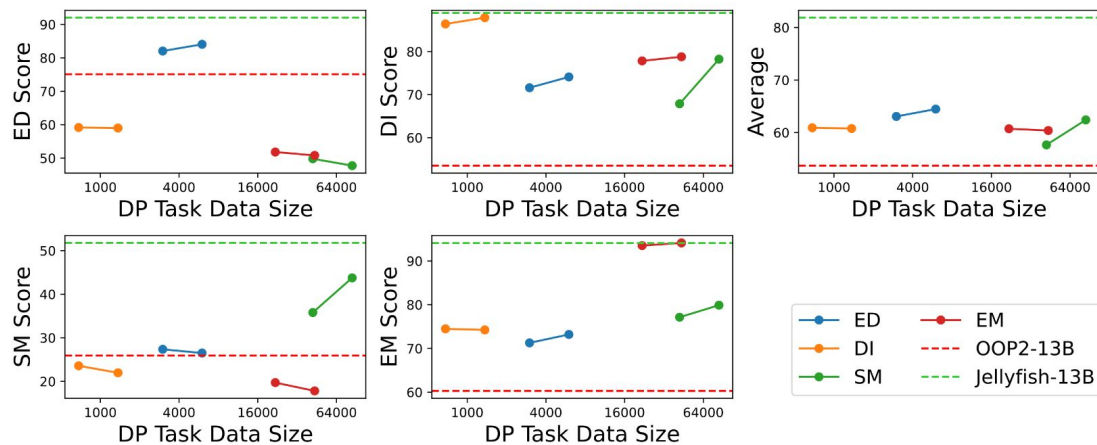


Figure 3: Impact of tuning with single-task data on DP performance, zero-shot. Above red line is positive.

Impact of Reasoning Data

- 7B and 8B Models
 - benefited from reasoning data
 - Optimal at 14k (7B) and 8k (8B)

- 13B Reasoning data drastically reduced DP performance, possibly because the base model's specialized reasoning/logic capabilities did not align well with DP logic

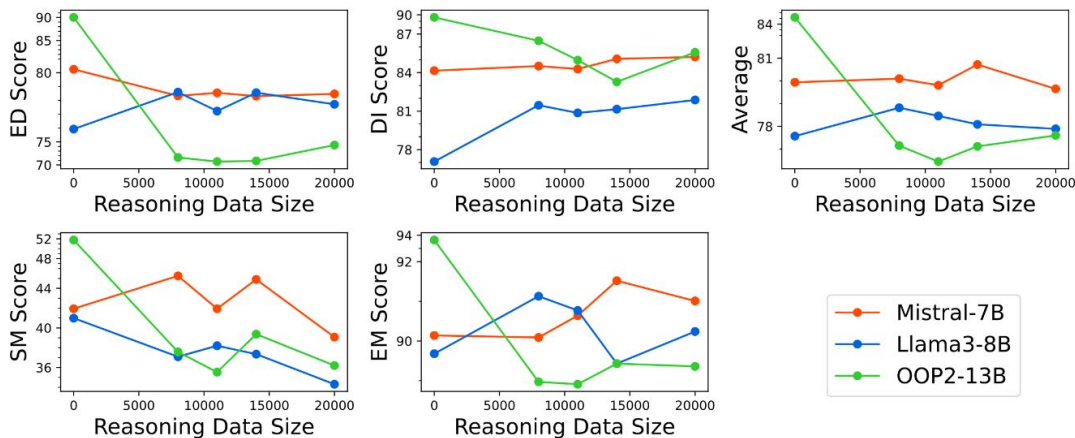


Figure 4: Impact of reasoning data on DP performance, zero-shot, plotted in log scale to show trends clearly.

Contributions

- First instruction-tuned LLMs for structured DP tasks
- Local, customizable, secure
- Enhanced reasoning and generalization

Encoding Spreadsheets for Large Language Models (SpreadsheetLLM)

Haoyu Dong, Jianbo Zhao, Yuzhang Tian, Junyu Xiong, Mengyu Zhou, Yun Lin,
José Cambronero, Yeye He, Shi Han, Dongmei Zhang

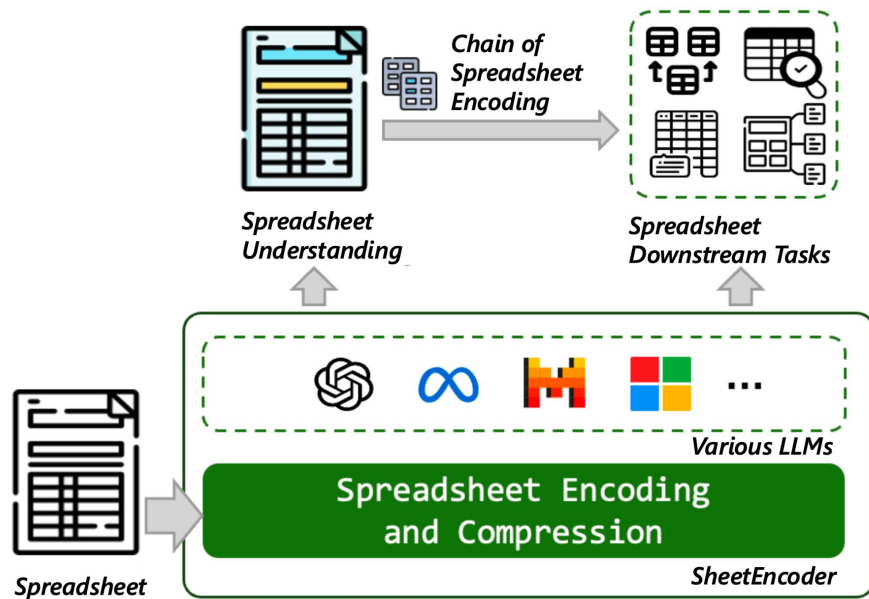
Microsoft Corporation
ACL 2024

Motivation & Problem

- Spreadsheets = large, 2D, flexible → difficult for LLMs
 - flexible layouts and **varied formatting**
 - **expansive** nature, too many tokens
 - struggle with linearizing **2D** layouts and processing spreadsheet-specific features (cell addresses and formats)
- Challenges: token limits, structure handling, format interpretation
- Introduce **Vanilla-encoding**, token consuming
 - basic serialization approach, including cell addresses, values, formats

Key Idea: SHEETENCODER

- Encoding + compression
- Improves **token efficiency** while retaining crucial **structural** and **semantic** information
- Wide range of downstream tasks
 - Chain of sheets (CoS)



Spreadsheet Example

	A	B	C	D	E	F	G	H
1	Name	Year	Country					
2		2009	Israel					
3	Ada Yonath	2011	United States			BIS	JAN	FEB
4		2010	Japan					
5		2007	United States			Girish	241	239
6						Joe	307	309
7						Amnon	308	297

Markdown Encoding

Name	Year	Country	...	...
Ada Yonath	2009	Israel	...	...
	2011	United States	...	BIS
	2010	Japan	...	
	2007	United States	...	Girish
			...	Joe
			...	Amnon

XML Encoding

HTML Encoding

```

<si>
  <t>Name</t>
</si>
<si>
  <t>Year</t>
</si>
<si>
  <t>Country</t>
</si>
<si>
  <t>United States</t>
</si>
...
<si>
  <t>JAN</t>
</si>
<si>
  <t>FEB</t>
</si>
<si>
  <t>Girish</t>
</si>
<si>
  <t>Joe</t>
</si>
<si>
  <t>Amnon</t>
</si>

```

```

<tr>
  <td>Name</td>
  <td>Year</td>
  <td>Country</td>
  <td></td>
  <td></td>
  <td></td>
  <td></td>
</tr>
<tr>
  <td>Ada Yonath</td>
  <td>2009</td>
  <td>Israel</td>
  <td></td>
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  <td></td>
  <td></td>
</tr>
<tr>
  <td></td>
  <td>2011</td>
  <td>United states</td>
  <td></td>
  <td></td>
  <td>BIS</td>
  <td>JAN</td>
  <td>FEB</td>
</tr>

```

Figure 5: Examples of three traditional spreadsheet encoding methods: Markdown, XML, and HTML. Due to space limitations, we only show the encoding of some cells.

Vanilla Encoding Strategy

- Vanilla encoding: addresses, values, formats
- **Markdown-like** style representation augmented with essential cell addresses and values
- Explored that **format** adversely affects model performance due to rapid token limit exceedance

$$\mathcal{S} = \{Cell_{i,j}\}_{i \in m, j \in n}, \quad (1)$$

$$\begin{aligned} \mathcal{T} &= \text{markdown} \{ \text{encode}(Cell_{i,j}) \} \\ &:= \text{"|Address}_{i,j}, Value_{i,j}, Format|... \backslash n", \end{aligned} \quad (2)$$

Text Input:

B2,Table 4: Diesel-driven passenger cars,
2015|C2, |D2, |E2, |F2, |G2, |H2,
B3, |C3, |D3, |E3, |F3, |G3, |H3,
B4, |C4, |D4, |E4, |F4, |G4, |H4,
|B5, |C5,Diesel engine|D5, |E5, |F5,Share of
all passenger cars (%)|G5, |H5,

.....

Format Input:

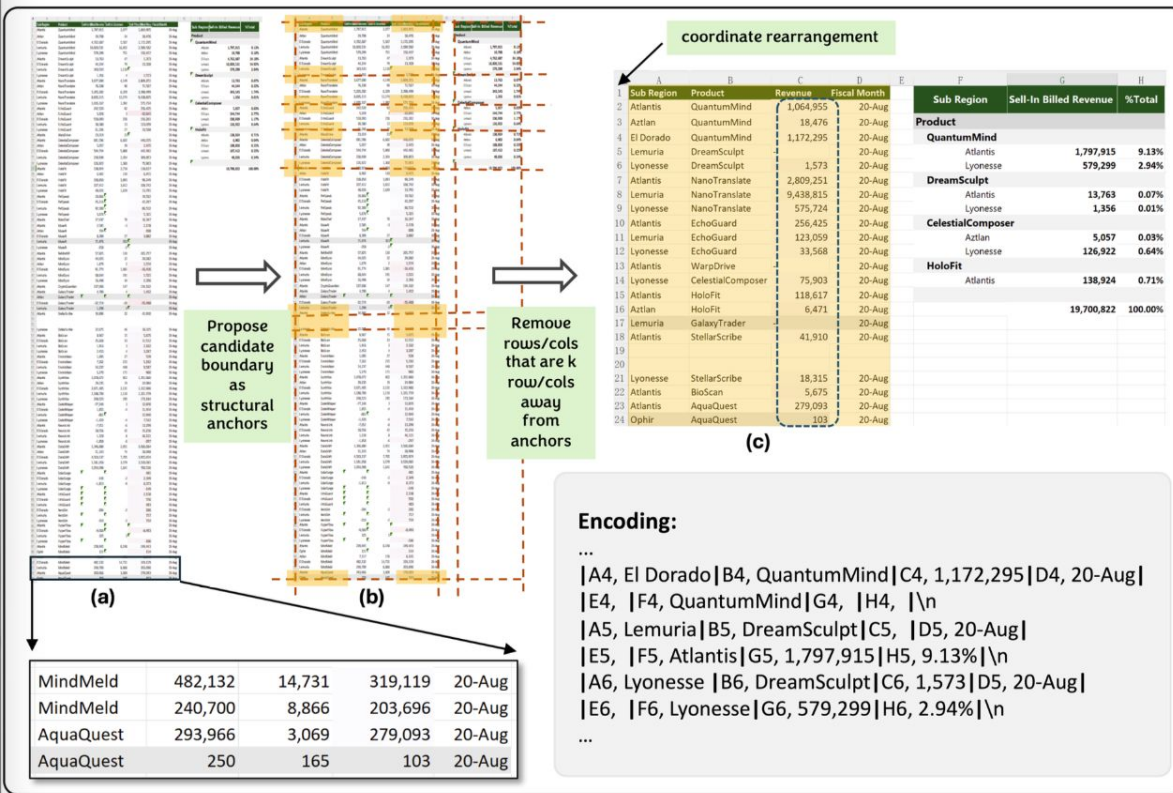
|B2,Font Bold|C2, |D2, |E2, |F2, |G2, |H2,
|B3, |C3, |D3, |E3, |F3, |G3, |H3,
|B4,Bottom Border,|C4,Bottom Bor-
der,|D4,Bottom Border,|E4,Bottom Bor-
der,|F4,Bottom Border,|G4,Bottom Bor-
der,|H4,Bottom Border,
|B5,Top Border,Right Border,Fill Color,Font
Bold|C5,Top Border,Bottom Border,Left
Border,Fill Color,Font Bold|D5,Top
Border,Bottom Border,Fill Color,Font
Bold|E5,Top Border,Bottom Border,Right
Border,Fill Color,Font Bold|F5,Top
Border,Bottom Border,Left Border,Fill
Color,Font Bold|G5,Top Border,Bottom
Border,Fill Color,Font Bold|H5,Top Bor-
der,Bottom Border,Fill Color,Font Bold

.....

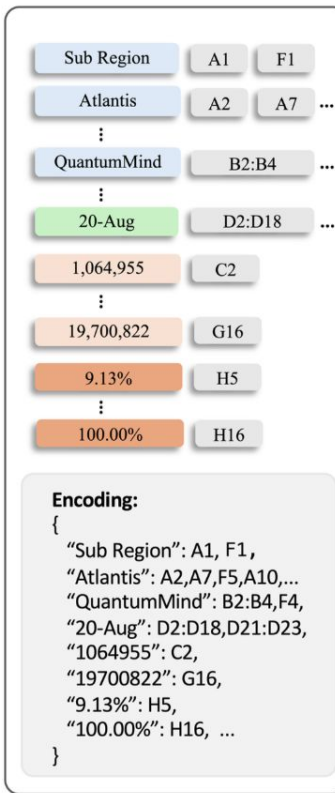
Compression Strategy

- Module 1: Structural-Anchor-Based Extraction
- Goal: Efficient layout understanding by focusing on heterogeneous areas
- Mechanism: Identifies "structural anchors"
 - rows and columns at possible table boundaries that offer substantial layout insights
 - rows/columns with differences/inconsistencies in values, merged cells, or borders
- Compression: Only keeps k-near, heterogeneous rows and columns,
 - producing a condensed "skeleton" version of the spreadsheet
 - k is set to 4, over 97% of the border rows and columns in ground truth tables are preserved

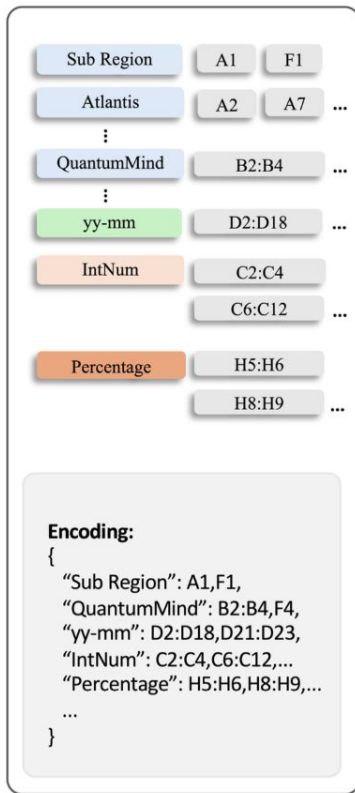
Illustration of the SHEETENCODER framework



1. Structural-anchor-based Extraction



2. Inverted-index Translation



3. Data-format-aware Aggregation

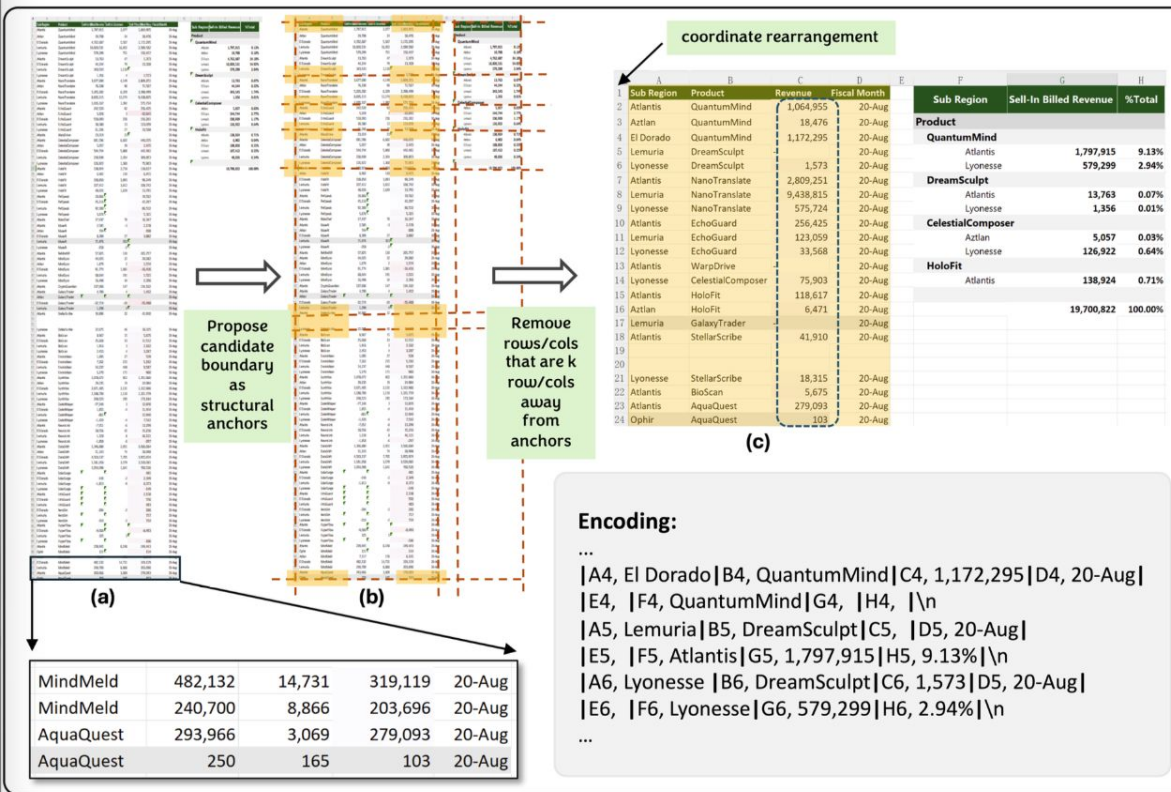
Compression Strategy

- Module 2: Inverted-Index Translation
- **lossless** compression method designed to eliminate redundancy from **empty and repeated cells**
- 2 stages
 - Convert the matrix-style encoding into a **dictionary** format
 - Cell values serve as keys indexing their addresses.
 - Cells sharing identical values are merged, and empty cells are excluded

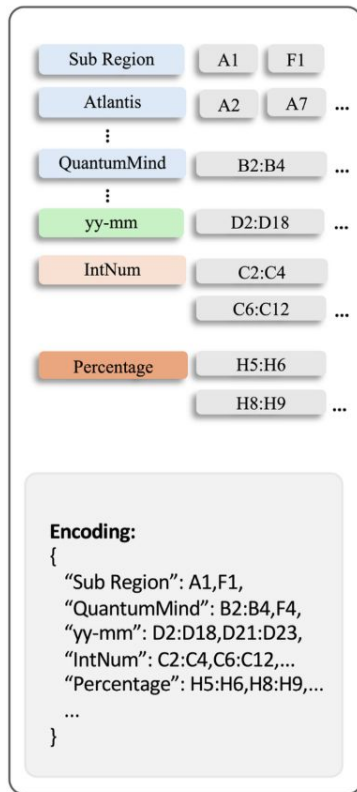
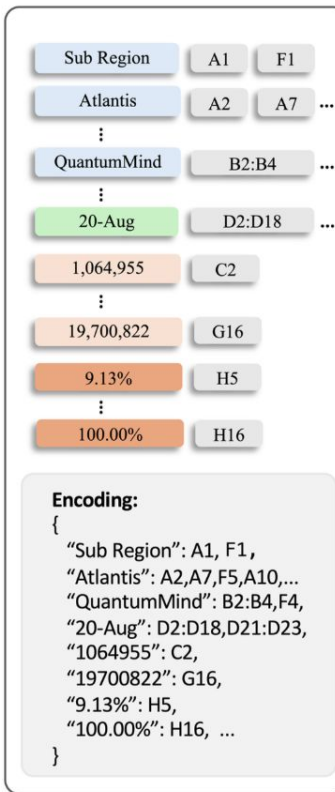
$$\mathcal{T}_t = \text{invert}(\mathcal{T})$$

$$:= \{Value : Address \text{ or } Address_Region, \dots\}.$$

Illustration of the SHEETENCODER framework



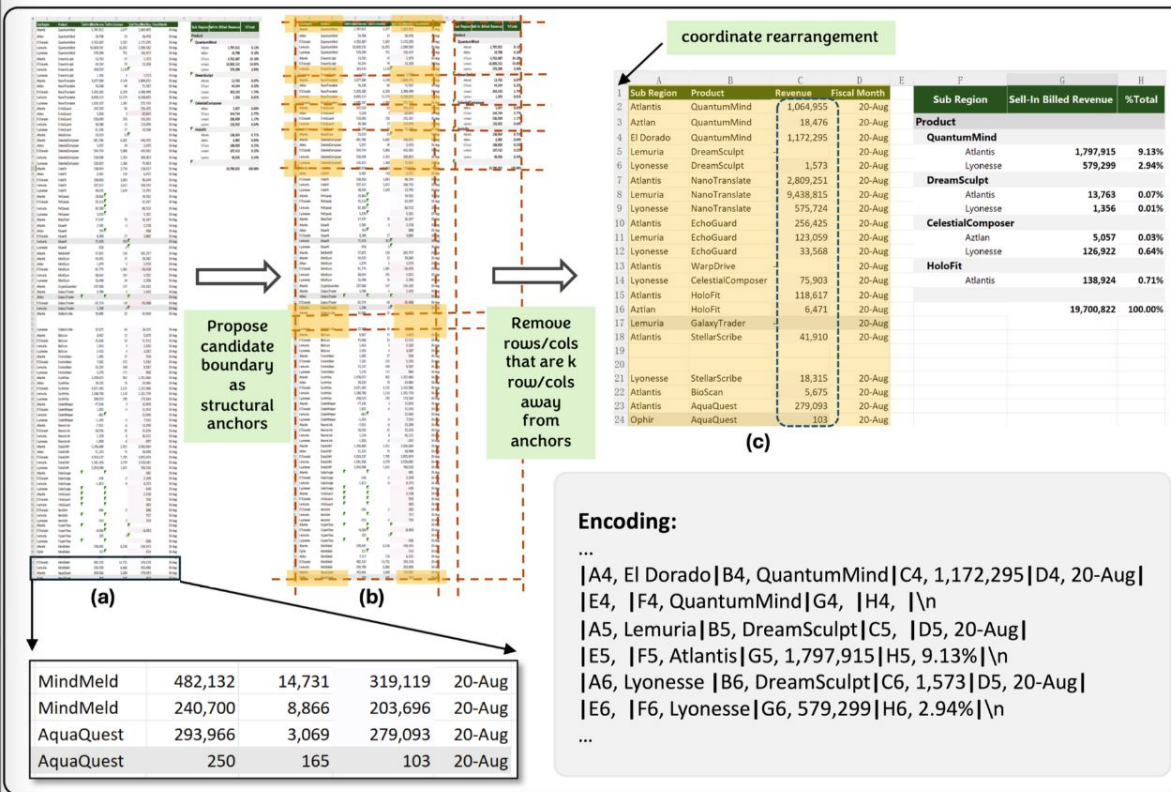
1. Structural-anchor-based Extraction



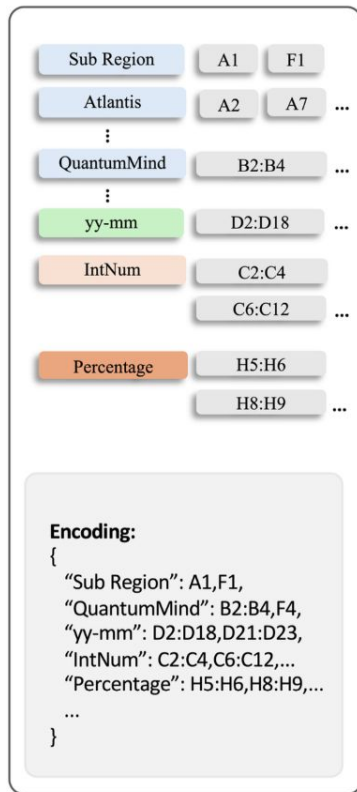
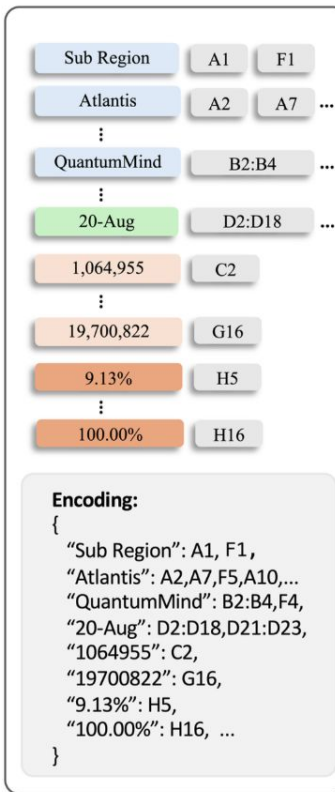
Compression Strategy

- Module 3: Data-Format-Aware Aggregation
- Hypothesis: **exact numerical** values are often less critical than structure and type
- Format/Type Recognition: Adjacent numerical cells are clustered based on:
 - Number Format String (NFS): **Built-in cell attributes** (e.g., "yyyy-mm-dd" for dates).
 - Rule-based Data Types: **Predefined types derived from rules** (e.g., Year, Integer, Float, Currency), covering about 55% of cells in the dataset
- Rectangular regions of cells sharing the same NFSs or data types are **aggregated**

Illustration of the SHEETENCODER framework



1. Structural-anchor-based Extraction



Chain of Sheets

- Multi-stage approach for downstream tasks (QA)
 1. **Table Identification**: The LLM processes the compressed spreadsheet and the query to identify the relevant table and its precise boundary.
 2. **Response Generation**: The query and the identified relevant table section (re-encoded) are fed back into the LLM to generate the final response.

Experiment Setup

- Tasks
 - Spreadsheet Table Detection (STD) (foundational)
 - Spreadsheet QA (downstream)
- STD Dataset:
 - 188 real-world spreadsheets (311 tables),
 - categorized by vanilla token size into Small, Medium, Large, and Huge (Huge > 32k tokens).
 - Evaluation used the Error-of-Boundary 0 (EoB-0) metric (exact boundary match) and Micro F1 Score

Experiment Setup

- QA Dataset: New multi-table QA dataset (307 items) where answers are formatted as **cell addresses or formulas**
- Models:
 - Closed-source (GPT-4, GPT-3.5; GPT-4 for QA)
 - Open-source (Llama2, Llama3, Phi3, Mistral-v2)
 - Tested via both In-Context Learning (**ICL**) and Fine-Tuning (**FT**)
 - TableSense-CNN: non-LLM
- Baseline QA models
 - TAPEX and Binder
 - Single-table data
 - Fine-tuned SpreadsheetLLM model identifies table regions

Compression Ratio Results

- Compression ratio: $r = n/n'$
- 25× on the test set, reducing token usage
- approximately 96% cost reduction

Metric	No Modules	Module 1	Module 2	Module 3	Module 1&2	Module 1&3	Module 2&3	Module 1&2&3
Total Tokens	1,548,577	350,946	580,912	213,890	103,880	96,365	211,445	62,469
Compression Ratio	1.00	4.41	2.67	7.24	14.91	16.07	7.32	24.79

Spreadsheet Table Detection Results

- Fine-tuned LLMs with compression
- compressed GPT-4 model (without aggregation)
- massive boosts on **larger** datasets

Model & Method	Small	Medium	Large	Huge	All
ICL					
Mistral-v2	0.071	0.013	0.029	0.017	0.036
GPT4	0.318	0.292	0.090	0.000	0.154
GPT4-compress	0.480	0.454	0.373	0.330	0.410
Fine-tune					
Llama3	0.715	0.765	0.290	0.000	0.471
Llama2	0.557	0.378	0.107	0.000	0.280
Phi3	0.604	0.481	0.201	0.130	0.330
Mistral-v2	0.700	0.784	0.472	0.123	0.542
GPT4	0.779	0.707	0.288	0.000	0.520
Llama3-compress	0.825	0.768	0.664	0.617	0.719
Llama2-compress	0.710	0.722	0.633	0.578	0.660
Phi3-compress	0.800	0.673	0.624	0.675	0.689
Mistral-v2-compress	0.778	0.729	0.686	0.744	0.726
GPT3.5-compress	0.795	0.649	0.600	0.680	0.680
GPT4-compress	0.810	0.832	0.718	0.690	0.759
-w/o Aggregation	0.864	0.816	0.739	0.753	0.789
TableSense-CNN	0.785	0.788	0.567	0.561	0.666

Ablation Studies in STD Results

- Impact of each compression module was evaluated by removing it from the full compressed configuration
- Removing module 1
 - largest performance drop (F1 fell from 0.759 to 0.655)
 - Critical
- Removing module 3
 - F1 score to slightly increase (from 0.759 to 0.789)
 - Challenges the LLMs' ability to interpret the data effectively

Model	Small	Medium	Large	Huge	All
GPT4	0.779	0.700	0.288	0.000	0.520
GPT4-compress	0.810	0.832	0.718	0.690	0.759
-w/o Extraction	0.805	0.772	0.618	0.321	0.655
-w/o Translation	0.785	0.804	0.729	0.636	0.743
-w/o Aggregation	0.864	0.816	0.739	0.753	0.789

Spreadsheet QA Results

- **Table-splitting** algorithm (rule-based) for tables exceptionally large and defy effective compression
- fine-tuned GPT-4 model combined with CoS and compression achieved 74.3% accuracy
- outperformed established Table QA baselines

Model	Accuracy
TAPEX	0.378
Binder	0.622
GPT4	0.466
GPT4-compress-w/o splitting	0.651
GPT4-compress-w/o splitting-FT	0.694
GPT4-compress	0.684
GPT4-compress-FT	0.743

Contributions

- First LLM-focused spreadsheet encoding framework
- Highly efficient token compression
- Strong reasoning on large/multi-table spreadsheets